Smart Clinic AI Assistant

System Analysis & Design

DEPI — Digital Egypt Pioneers Initiative | Track: AI & Data Science

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Repository	https://github.com/nhahub/NHA-5-201
Version / date	1.0 — 6 November 2026

1. Problem Statement & Objectives

Problem. A small clinic's secretary spends the day answering the same questions (address, fees, working hours), booking appointments by hand, chasing patients who forget their appointments, and adding up payments at night. This causes double-booked slots, wrong totals, unanswered messages outside working hours, and no structured view of how patients feel about the service.

Proposed solution. An automation system built on n8n and delivered through a Telegram bot. Patients choose from button menus, book appointments against real Google Calendar availability, and receive reminders. Scheduled workflows produce the daily financial report and collect post-visit feedback. An AI agent answers free-text questions, restricted to clinic information.

Design principle. Anything that must be exact (availability, bookings, money) is decided by code. The LLM is used only to understand and produce language, and never decides a booking or a number.

Objectives

- Answer recurring patient questions on Telegram around the clock.
- Let patients self-book with real-time availability checks, with no double-booking.
- Reduce forgotten appointments with an interactive reminder 24 hours before the visit (Confirm / Reschedule / Cancel).
- Give the doctor an accurate automated daily financial report (income and case count).
- Measure patient satisfaction after each visit by classifying feedback as Positive, Neutral, or Negative.

Scope. A single-doctor clinic, 20-minute appointments, working hours 09:00 to 15:00 (Africa/Cairo), Friday and Saturday off, Telegram as the patient channel.

2. Use Case Diagram & Descriptions

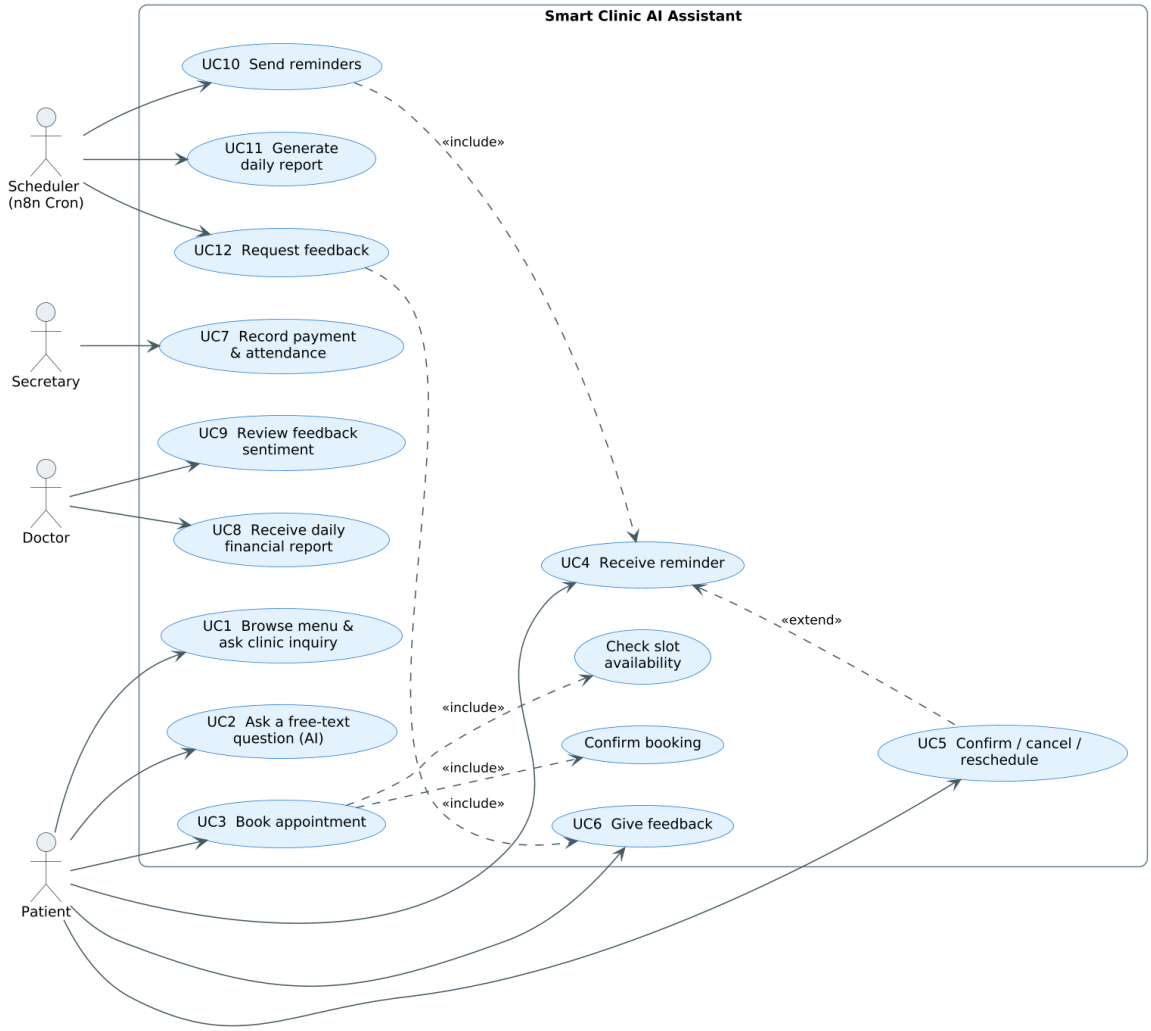


Figure 1 — Use case diagram

Actors

Actor	Role
Patient	Uses the Telegram bot to ask questions, book, respond to reminders, and give feedback.
Secretary	Records payments and attendance in the ledger sheet.
Doctor	Receives the daily financial report and reviews feedback results.
Scheduler (n8n Cron)	Automated actor that triggers reminders, the daily report, and feedback requests.

Use case descriptions

ID	Use case	Actor	Main flow	Alternative / exception
UC1	Browse menu and ask a clinic inquiry	Patient	Sends /start, sees the menu, taps Inquiry, receives address, fees, and working hours.	Types free text instead: see UC2.
UC2	Ask a free-text question	Patient	Types a question outside the button flow. The AI Agent answers using clinic information only and points the patient to /start for booking.	AI service unavailable: the bot shows the main menu.

ID	Use case	Actor	Main flow	Alternative / exception
UC3	Book an appointment	Patient	Taps Book, picks a day, picks a free slot, sends name and age, reviews the summary, taps Confirm. The slot is re-checked, a Calendar event is created, and the booking is logged to Sheets.	Invalid name or age: asked again. Slot taken meanwhile: told to choose another. Cancel: back to the menu.
UC4/5	Receive a reminder and respond	Patient	Receives a reminder 24 hours before. Taps Confirm, Reschedule, or Cancel. The booking status is updated.	No response: status unchanged. Cancel: the Calendar event is deleted.
UC6/12	Give feedback	Scheduler, Patient	After an attended visit the bot asks for feedback. The text is classified and stored.	No reply: nothing is stored.
UC7	Record payment and attendance	Secretary	Enters the amount and marks attendance in the sheet.	—
UC8/11	Daily financial report	Scheduler, Doctor	At end of day code totals income and counts cases from the ledger. The report is emailed to the doctor.	No rows today: report shows zero.
UC9	Review feedback	Doctor	Opens the Feedback sheet and reads text and sentiment.	—
UC10	Send reminders	Scheduler	Finds bookings due in the next 24 hours that were not reminded, sends the reminder, sets reminder_sent.	Reminder already sent: skipped.

3. Functional & Non-Functional Requirements

This section summarizes the requirements from the Requirements Gathering document, updated for the AI agent and the conversation-state fix.

Functional requirements

ID	Requirement
FR1	The system shall respond to patients on Telegram 24/7.
FR2	The main menu and all booking steps shall be button-driven. Free-text messages that are not part of an active booking step shall be answered by the AI Agent.
FR3	Inquiry answers (address, fees, working hours) shall come from fixed clinic data. The AI Agent shall answer only from clinic information, shall not book appointments, and shall not give medical advice.
FR4	Availability shall be computed by code from Google Calendar: 20-minute slots between 09:00 and 15:00 (Africa/Cairo), Friday and Saturday excluded, past and overlapping slots removed.
FR5	The chosen slot shall be re-checked against the calendar at confirmation, before the event is created.
FR6	The system shall collect the patient's name (at least three words) and age (1 to 110) and show a summary for confirmation. Phone number is planned.
FR7	The system shall remember the chosen slot per patient (chat_id) so name and age can be sent as a normal message or as a reply. The state expires after 30 minutes and is cleared by /start or any menu button.
FR8	Each confirmed booking shall be logged to Google Sheets with a unique id, chat_id, date, time, and status.
FR9	A reminder with Confirm / Reschedule / Cancel buttons shall be sent about 24 hours before each appointment, once only. Cancel deletes the Calendar event.
FR10	Cancellation or rescheduling shall be allowed up to 3 hours before the appointment (default assumption, pending confirmation with the client).
FR11	The daily total income and case count shall be calculated by code (not by the LLM) and emailed to the doctor.

ID	Requirement
FR12	Feedback shall be requested only for attended visits and classified as Positive, Neutral, or Negative by an LLM, then stored with the original text.
FR13	A patient shall only be able to act on their own bookings (matched by chat_id).

Non-functional requirements

Category	Requirement
Performance	Button-driven replies should arrive within a few seconds. AI replies may take longer and are only used for free text.
Reliability	Money and availability are deterministic (code). If the AI service fails, the bot falls back to the menu.
Security	Tokens and OAuth credentials live only in n8n credentials and are never committed to the repository. Each patient can act only on their own booking.
Privacy	Only name and age are collected. No medical details are requested or stored.
AI safety	The AI Agent has no booking tools, a restricted system prompt, and redirects medical questions to the clinic.
Usability	Short Arabic messages, buttons instead of typing, and a clear way out (/start, Back, Cancel) at every step.
Maintainability	Workflows are exported as JSON to the repository (without credentials), with descriptive node names, one workflow per part.
Availability	Depends on the n8n hosting (n8n Cloud or self-hosted with a public HTTPS URL).

4. Software Architecture

Architecture style. Event-driven workflow orchestration in four layers. Telegram updates and scheduled timers trigger n8n workflows. Deterministic Code nodes make every decision that must be exact, and an LLM is used only where language understanding is needed. Google services provide the data layer.

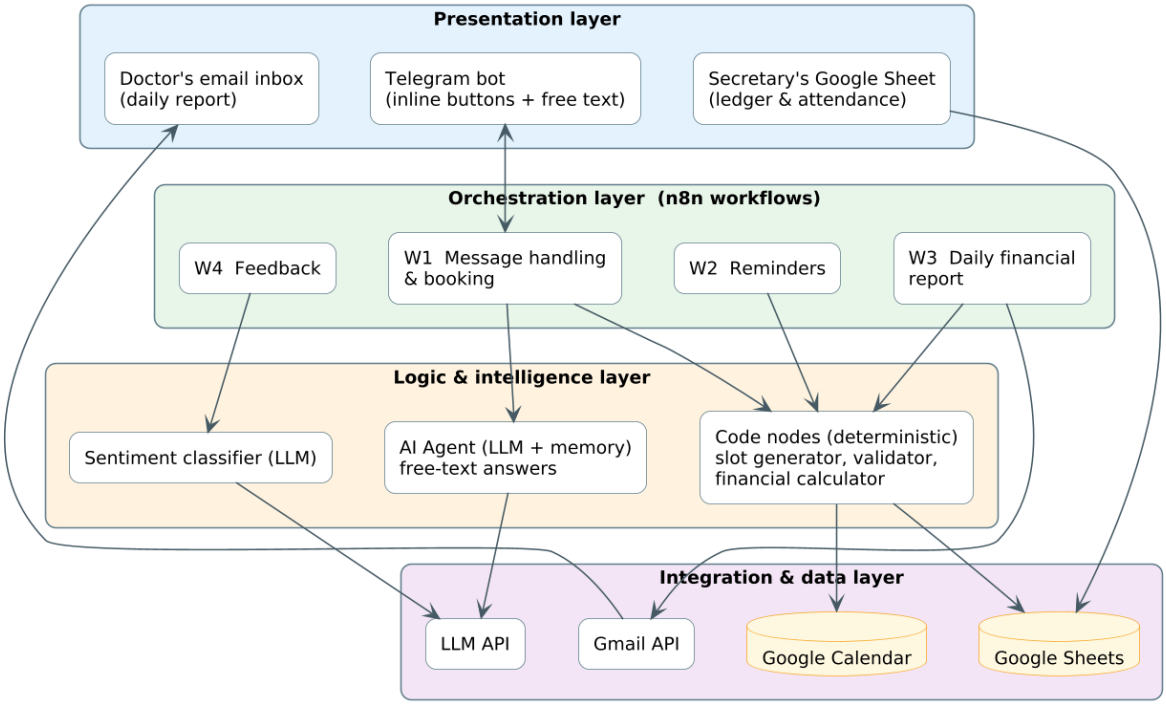


Figure 2 — Layered architecture

Decision	Reason
Buttons first, AI for free text	Buttons are faster, cheaper, and exact. The AI covers the cases buttons cannot, without touching bookings.
Code decides, LLM understands	Prevents hallucinated slots or totals. Availability, booking, and money never depend on an LLM.
Google Calendar as the source of truth for availability	The doctor sees appointments on their phone, and walk-in bookings made directly in the calendar are respected.
Google Sheets as the MVP data store	No setup, easy for the secretary to read and edit. The logical model is relational so it can move to PostgreSQL later.
Conversation state in n8n workflow static data	Simple and enough for one clinic. It can move to a Sessions sheet or a database when the system grows.
Two separate LLM uses	The AI Agent (free-text answers) and the sentiment classifier (feedback) are separate, each with a narrow prompt.

5. Database Design & Data Modeling

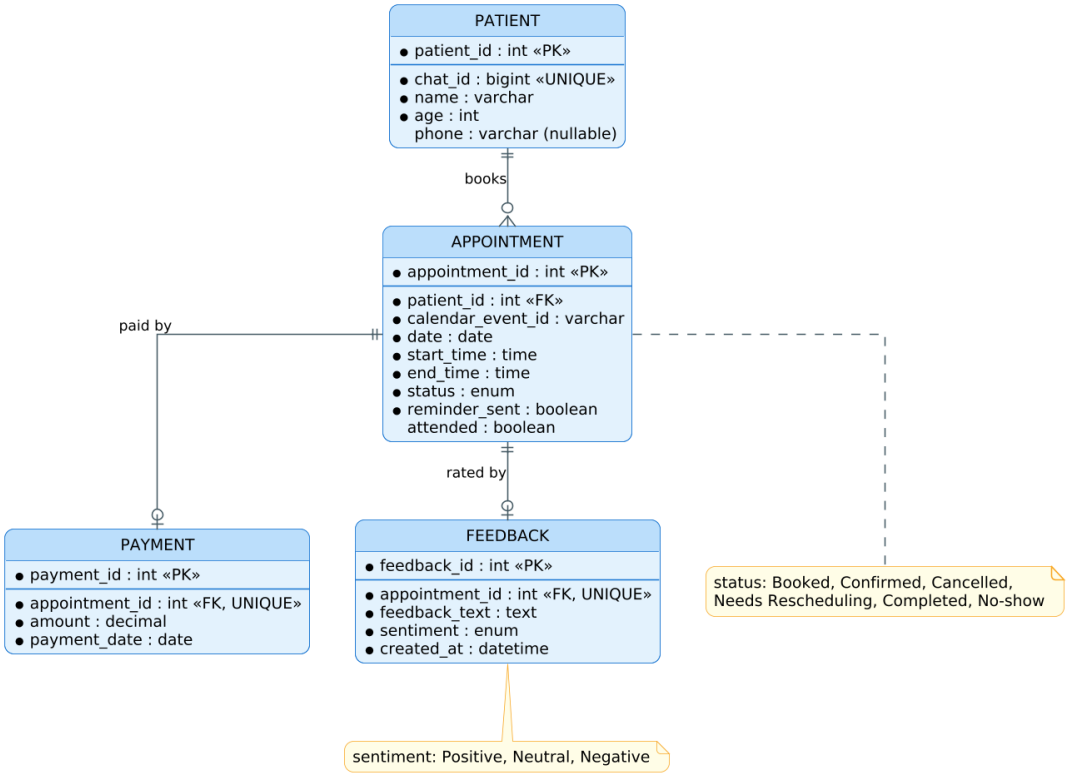


Figure 3 — Entity-relationship diagram

Logical schema (third normal form)

Entity	Attributes	Keys and constraints
PATIENT	patient_id, chat_id, name, age, phone (nullable)	PK patient_id. chat_id unique and not null. name and age not null.
APPOINTMENT	appointment_id, patient_id, calendar_event_id, date, start_time, end_time, status, reminder_sent, attended	PK appointment_id. FK patient_id references PATIENT. status in {Booked, Confirmed, Cancelled, Needs Rescheduling, Completed, No-show}. No two active appointments with the same date and start_time.

Entity	Attributes	Keys and constraints
PAYMENT	payment_id, appointment_id, amount, payment_date	PK payment_id. FK appointment_id references APPOINTMENT, unique (one payment per appointment). amount greater than or equal to 0.
FEEDBACK	feedback_id, appointment_id, feedback_text, sentiment, created_at	PK feedback_id. FK appointment_id references APPOINTMENT, unique. sentiment in {Positive, Neutral, Negative}.

Relationships: one patient has many appointments; an appointment has at most one payment and at most one feedback. There is no Doctor table because the MVP has a single doctor; clinic settings (hours, slot length) live in the workflow code. The conversation state (chat_id, pending date and time, created time) is transient and stored in n8n workflow static data, not in the data store.

Physical schema (Google Sheets, MVP)

Sheet	Columns	Notes
Bookings	appointment_id, chat_id, name, age, phone, date, start_time, end_time, status, reminder_sent, attended, calendar_event_id	Patient and Appointment are merged into one sheet for simplicity. Patient columns repeat per booking (accepted denormalization).
Daily Ledger	date, appointment_id, patient_name, amount	Filled by the secretary. The daily report reads today's rows.
Feedback	appointment_id, patient_name, feedback_text, sentiment, created_at	Filled by the feedback workflow.

Normalization and trade-offs. The logical model is in third normal form: patient attributes are stored once, and payment and feedback depend only on the appointment. The physical MVP deliberately denormalizes Bookings because Google Sheets has no foreign keys, joins, or transactions. If volume or concurrency grows, the logical schema maps directly to PostgreSQL tables.

6. Data Flow & System Behavior

6.1 Data Flow Diagrams

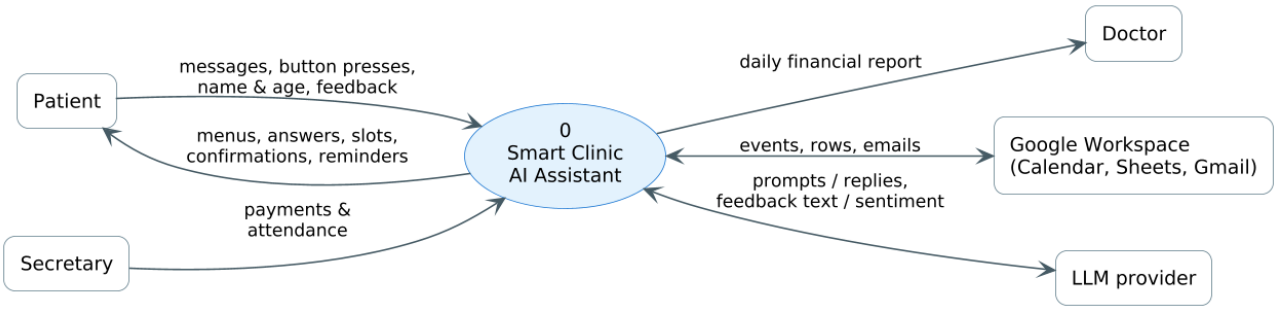


Figure 4 — DFD level 0 (context diagram)

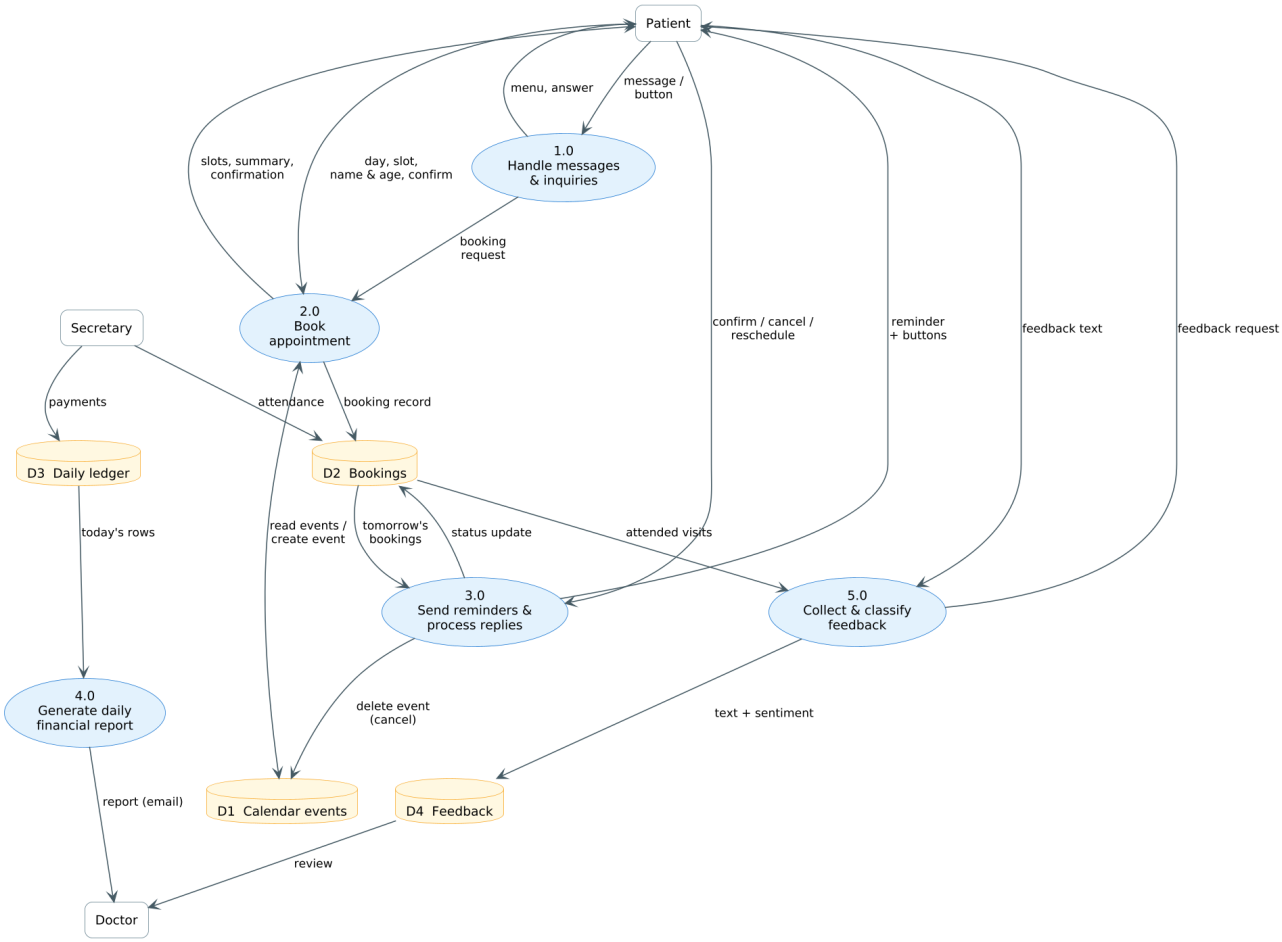


Figure 5 — DFD level 1

Process / store	Description
1.0 Handle messages and inquiries	Routes each update: menu, inquiry answers, or the AI Agent for free text.
2.0 Book appointment	Day and slot selection, validation, re-check, event creation, booking record.
3.0 Send reminders and process replies	Reads upcoming bookings, sends reminders, applies Confirm, Cancel, or Reschedule.
4.0 Generate daily financial report	Totals the day's ledger rows in code and emails the doctor.
5.0 Collect and classify feedback	Requests feedback for attended visits and stores text with sentiment.

Process / store	Description
D1 to D4	Calendar events, Bookings sheet, Daily Ledger sheet, Feedback sheet.

6.2 Sequence Diagrams

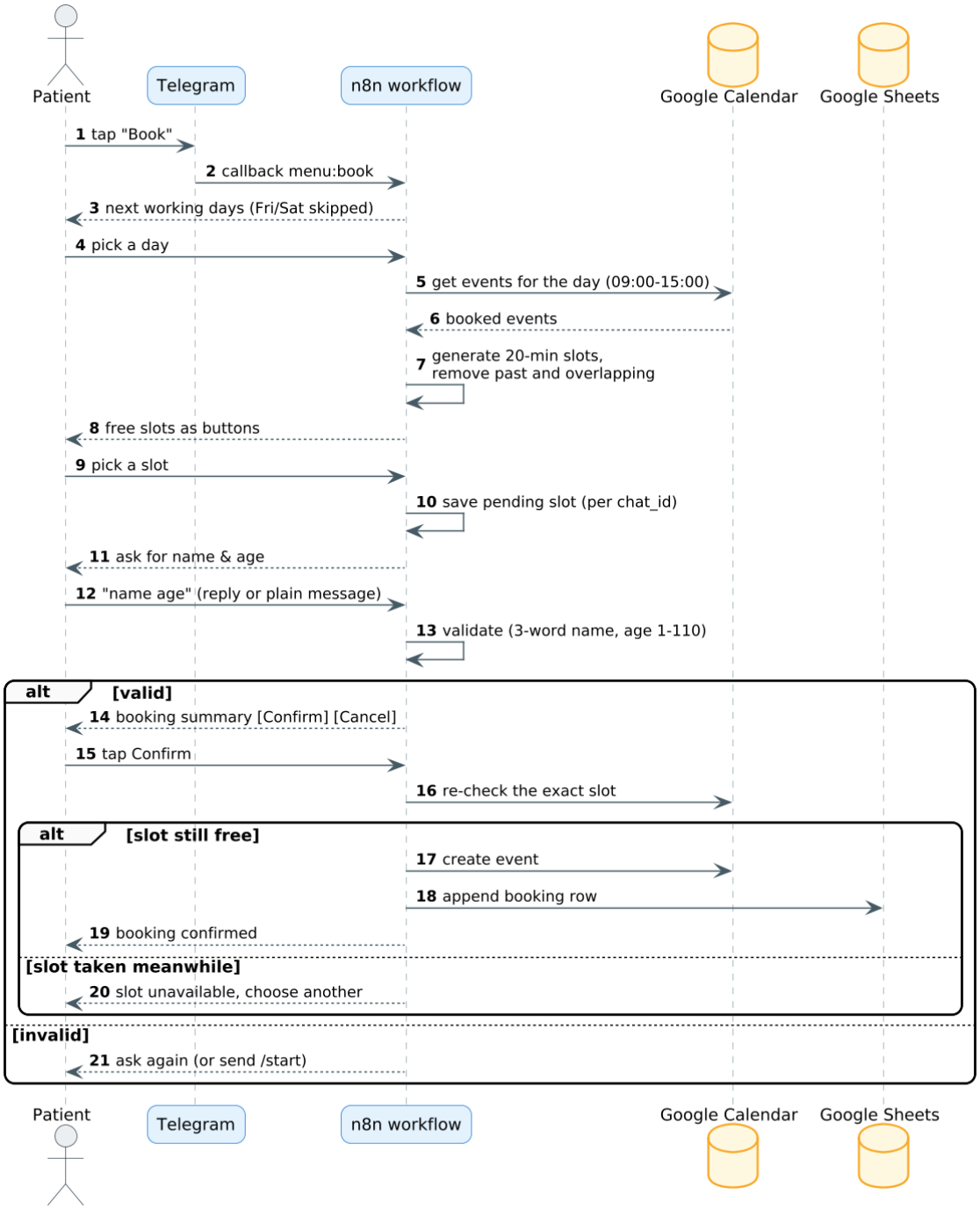


Figure 6 — Booking flow

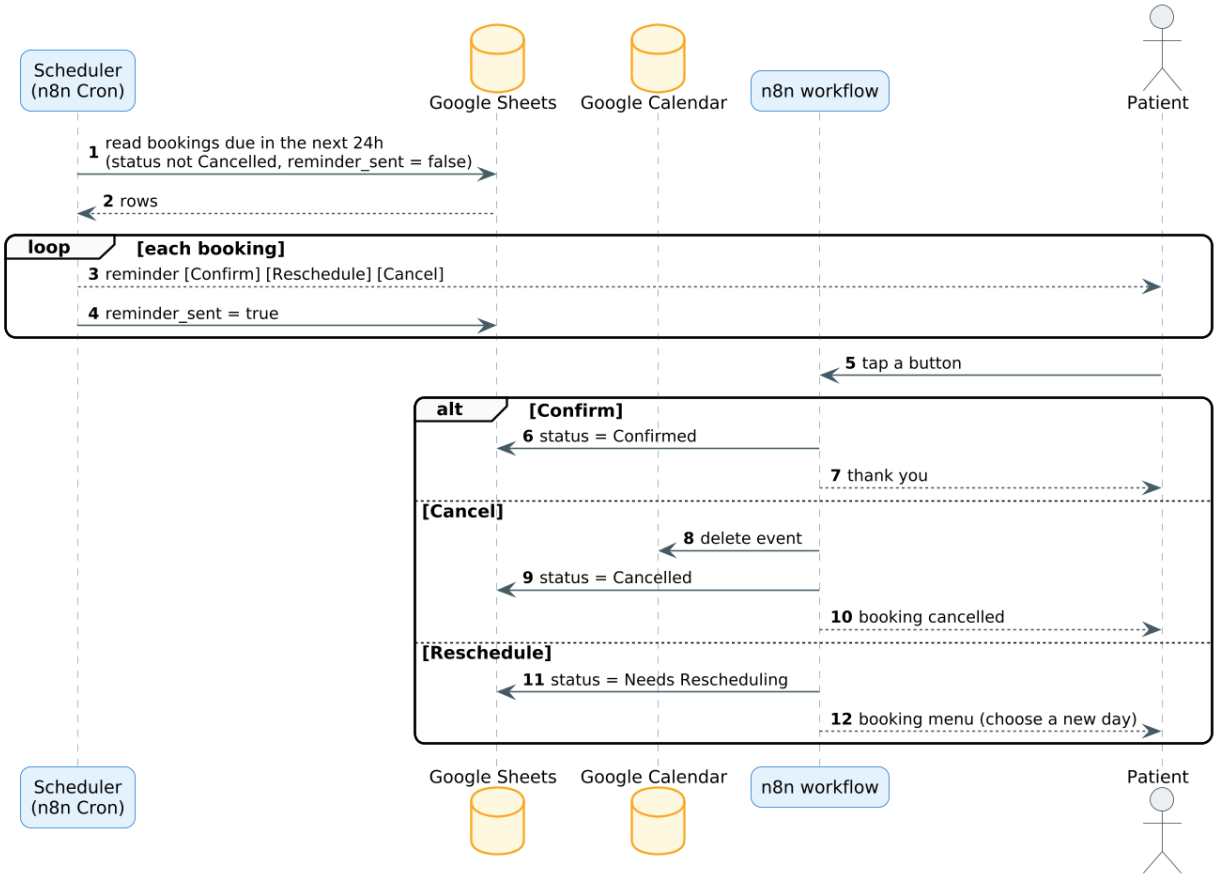


Figure 7 — Reminder flow

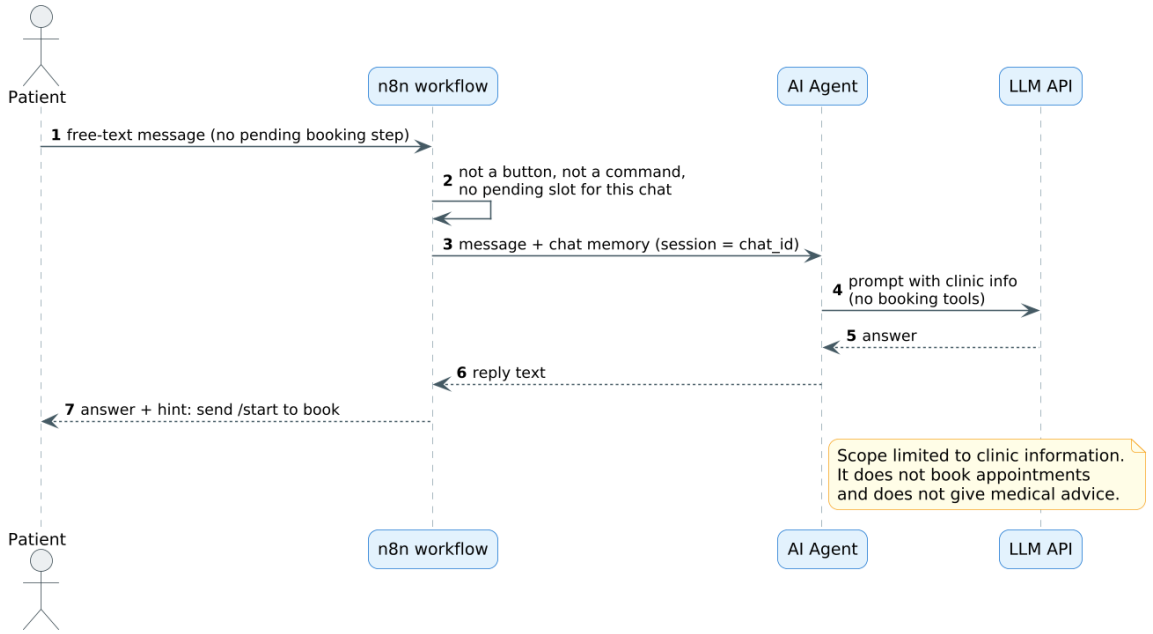


Figure 8 — Free-text question answered by the AI Agent

6.3 Activity Diagrams

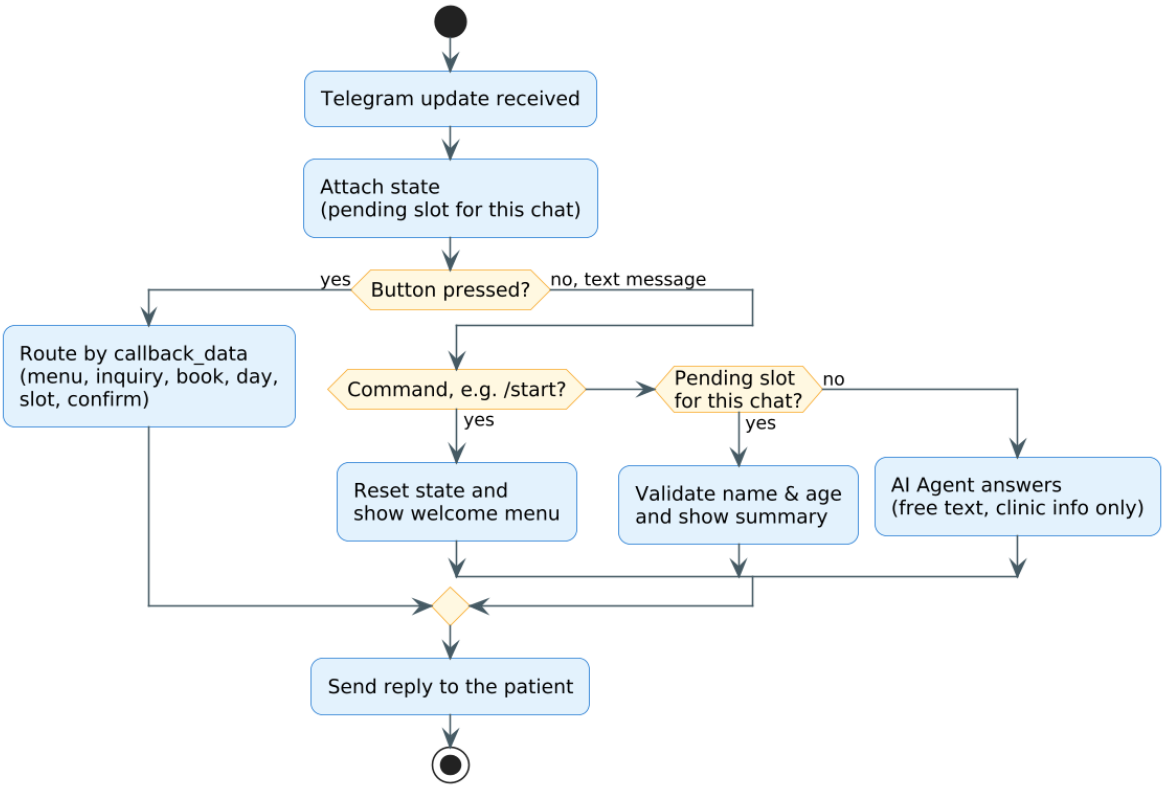


Figure 9 — Message routing

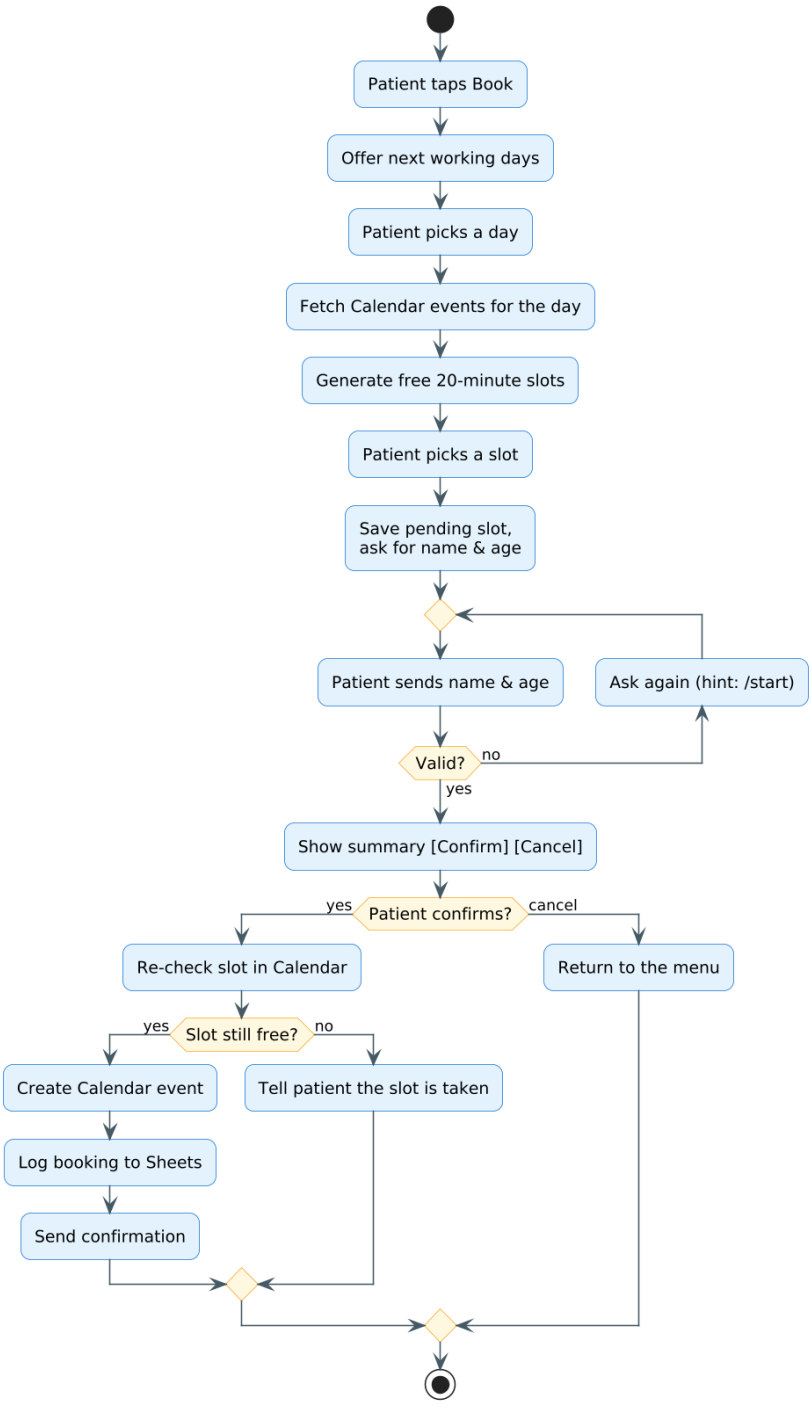


Figure 10 — Booking flow

6.4 State Diagram

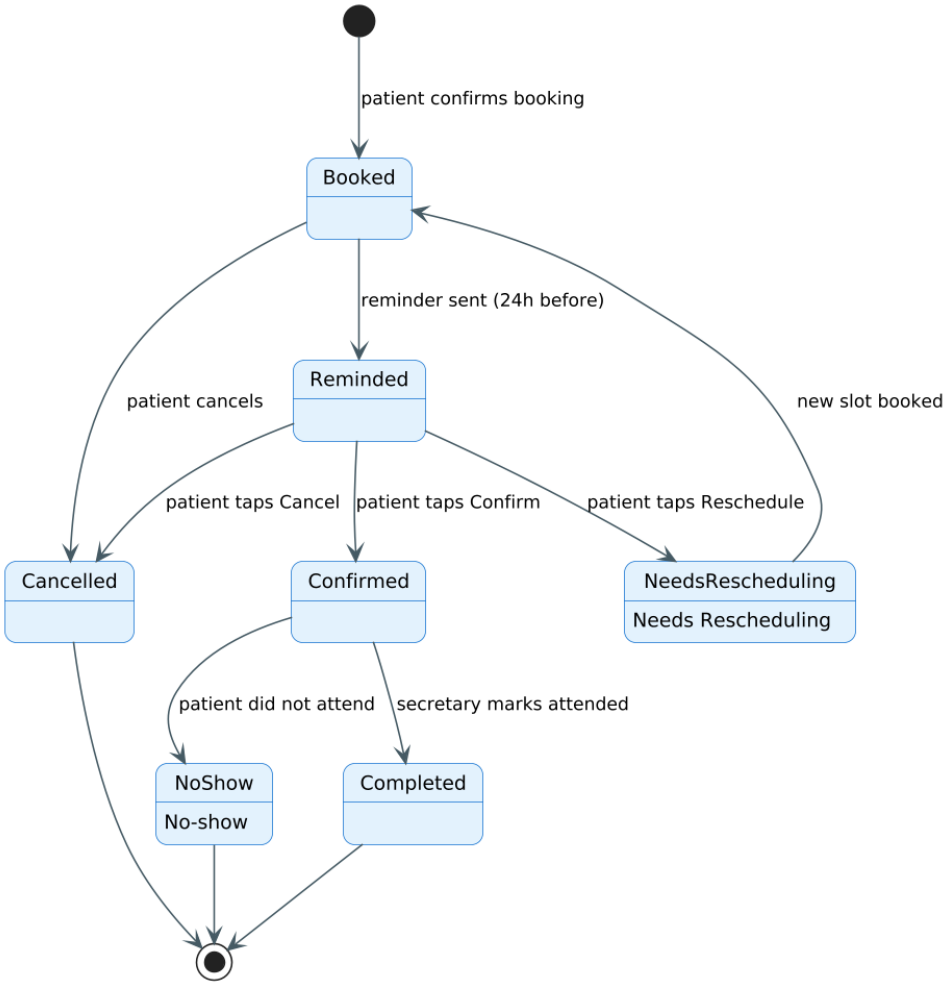


Figure 11 — Appointment states

State	Meaning	Triggered by
Booked	Slot reserved and event created.	Patient confirms the booking.
Reminded	Reminder sent 24 hours before.	Scheduler.
Confirmed	Patient confirmed attendance.	Confirm button.
Needs Rescheduling	Patient asked to move the appointment.	Reschedule button, then a new booking.
Cancelled	Appointment cancelled, Calendar event deleted.	Cancel button.
Completed / No-show	Visit happened or not.	Secretary marks attendance.

6.5 Class Diagram

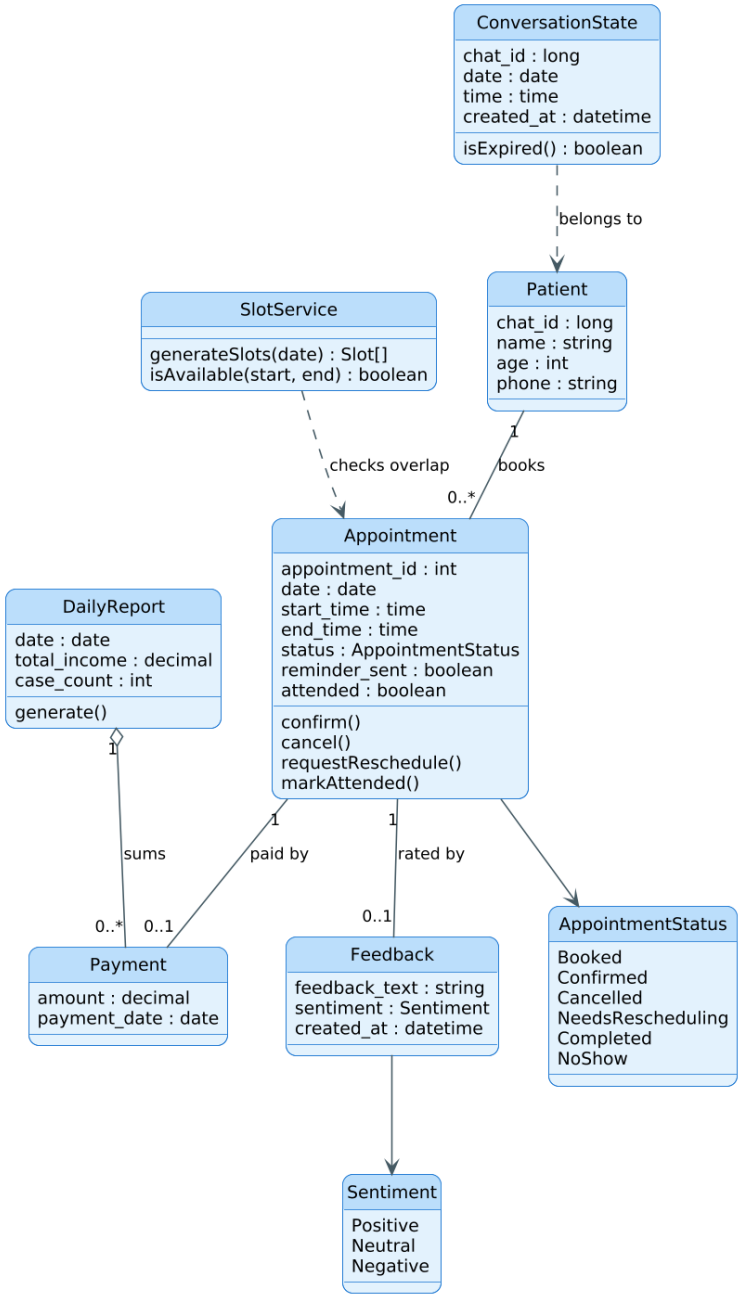
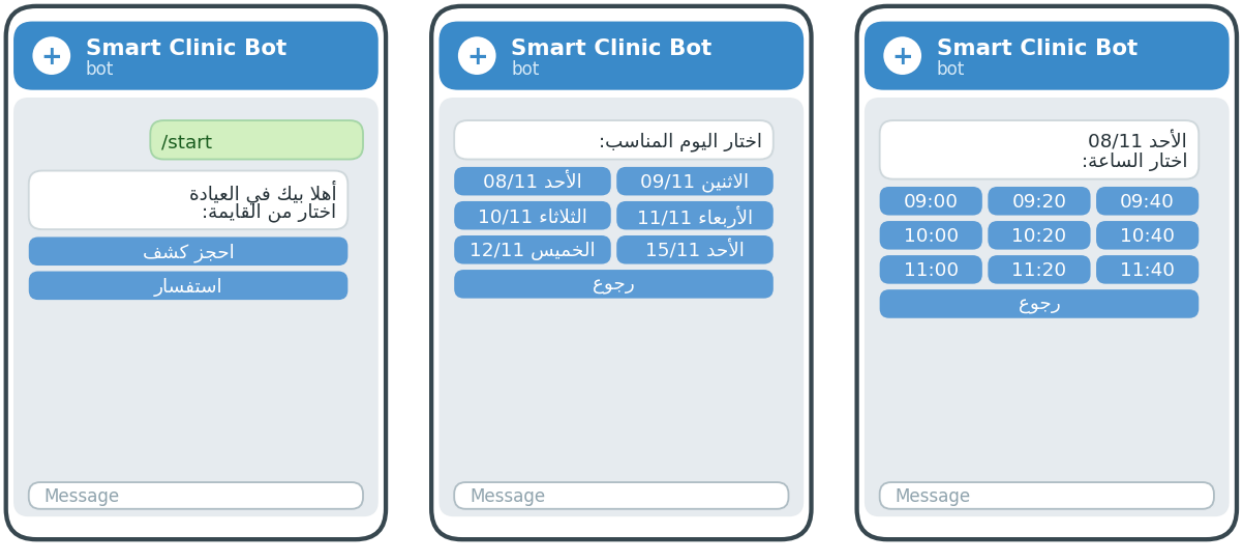


Figure 12 — Domain classes (logical model)

The classes describe the logical domain. In the MVP they are realized as n8n Code nodes and Google Sheets rows rather than as code classes: SlotService corresponds to the slot-generation Code nodes, and ConversationState to the workflow static data.

7. UI/UX Design & Prototyping

The interface is the Telegram chat itself, so the wireframes show conversation screens. Colors, fonts, and dark mode are inherited from the patient's Telegram app. Messages are in Egyptian-friendly Arabic.

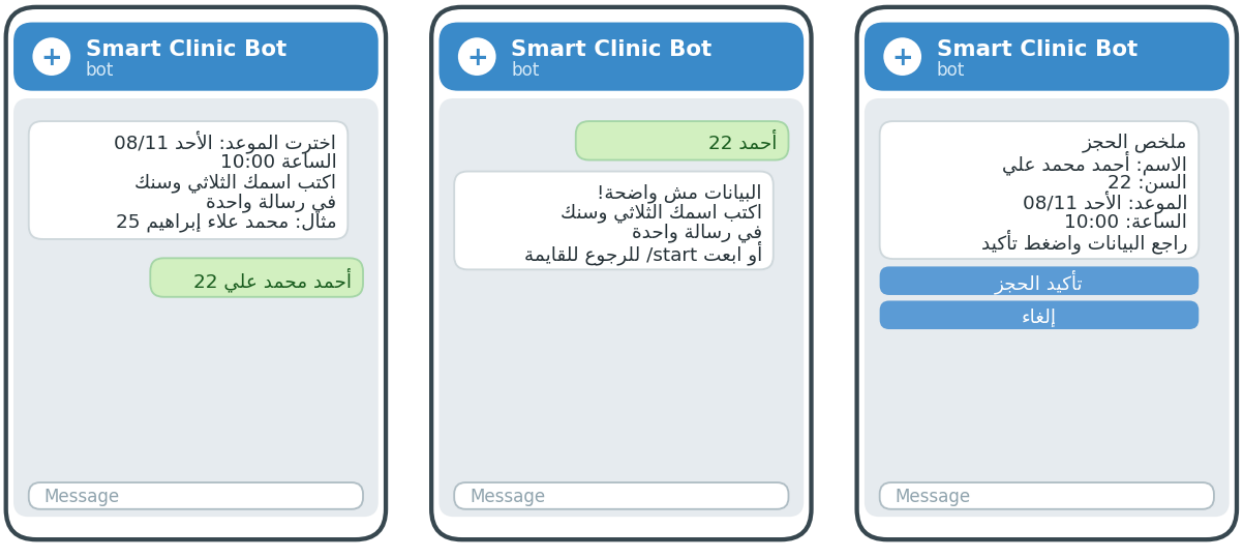


1. Main menu
Button-driven entry point

2. Choose a day
Next 6 working days (Fri/Sat skipped)

3. Choose a time
Only free 20-minute slots are shown

Figure 13 — Wireframes 1 to 3: menu, day, and time selection



4. Name and age
Works as a reply or as a plain message

5. Invalid input
Clear error and a way out

6. Booking summary
Confirmation before anything is saved

Figure 14 — Wireframes 4 to 6: name and age, invalid input, booking summary

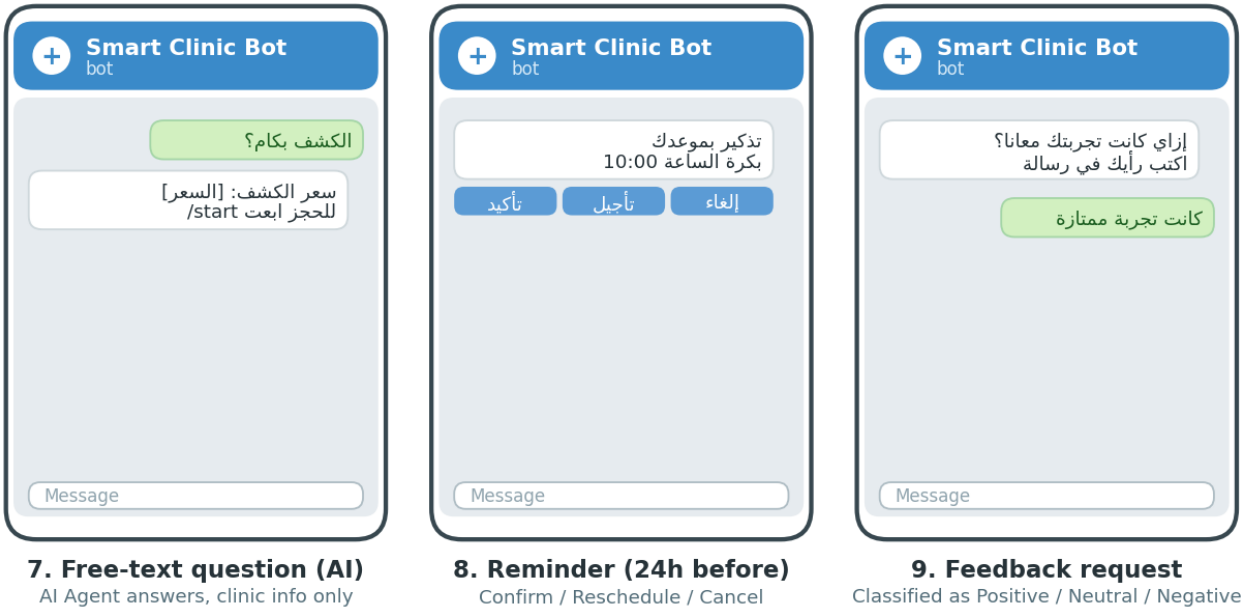


Figure 15 — Wireframes 7 to 9: AI answer, reminder, feedback request

UI/UX guidelines

- **Buttons first.** Choices are inline buttons. Typing is needed only for name and age, free-text questions, and feedback.
- **Compact layout.** At most two columns for days and three for times; Back is always the last row.
- **Always a way out.** /start, Back, and Cancel are available at every step, and error messages mention /start.
- **Confirm before saving.** A summary with Confirm and Cancel appears before any booking is created.
- **Helpful errors.** Say what is wrong and show an example of the expected input.
- **Consistent language.** Short sentences, the same wording for confirmations and cancellations, and sparing use of emojis.
- **Accessibility.** Every button has a text label (no icon-only buttons), and the bot relies on Telegram's font scaling, screen reader support, right-to-left text, and dark mode.
- **Privacy by design.** Only name and age are requested. The AI Agent does not give medical advice.

8. System Deployment & Integration

Technology stack

Layer	Technology
Frontend (patient interface)	Telegram chat: inline-button menus and free text
Backend (logic and orchestration)	n8n workflows with JavaScript Code nodes (including luxon for dates and time zones)
Database / data store	Google Calendar (appointments) and Google Sheets (bookings, ledger, feedback)
AI	LLM API used by an n8n AI Agent (free-text answers) and by the sentiment classifier
Notifications	Telegram Bot API (messages) and Gmail API (daily report)
Source control and documentation	GitHub repository: documents, workflow JSON exports, README

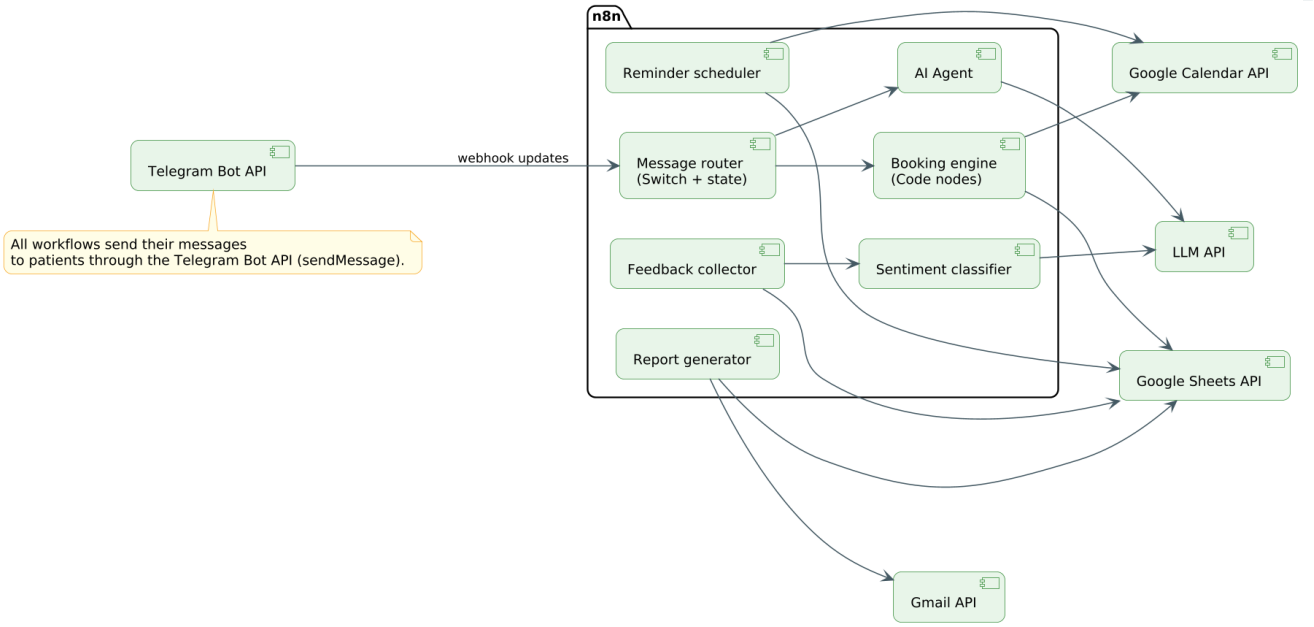


Figure 16 — Component diagram

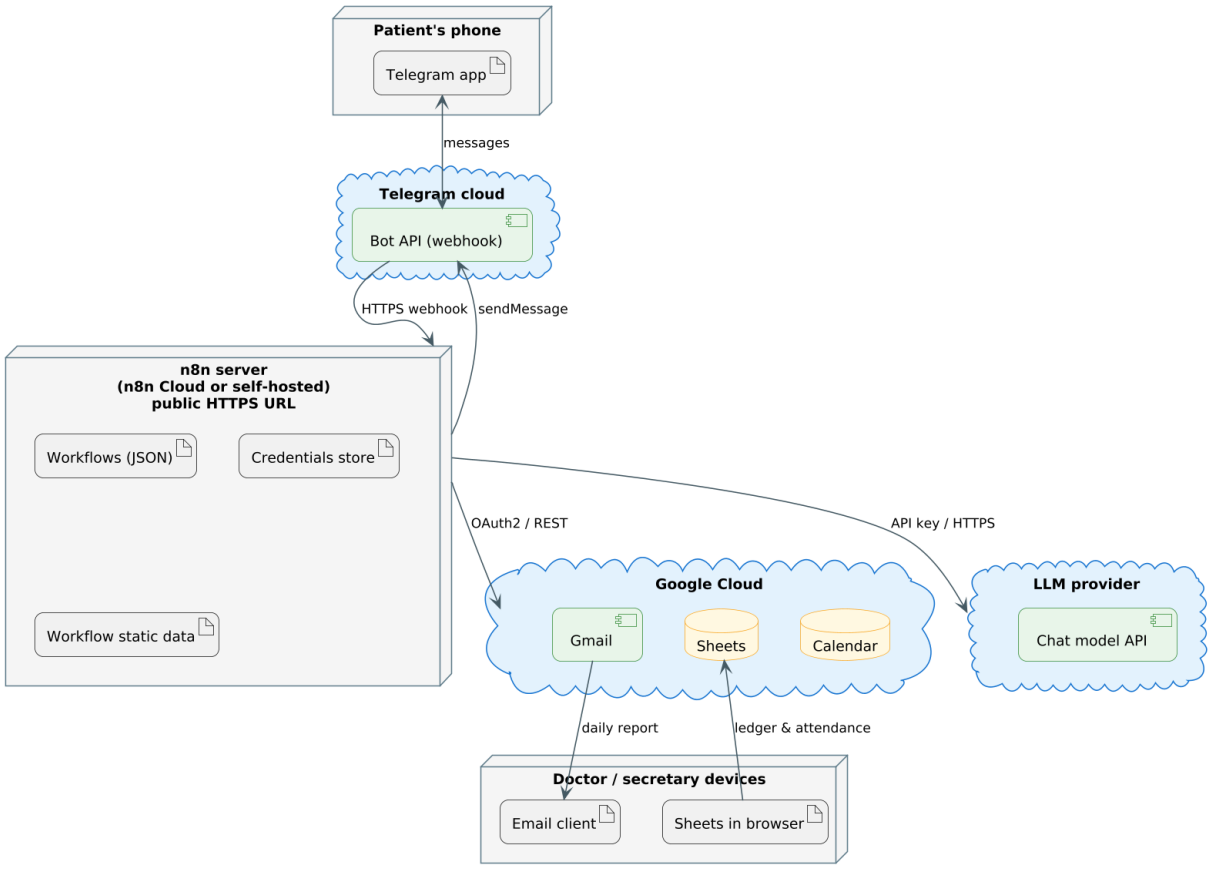


Figure 17 — Deployment diagram

Integration notes. Telegram reaches n8n through an HTTPS webhook, so n8n needs a public HTTPS URL (n8n Cloud, or a self-hosted instance behind a tunnel or reverse proxy). Google services use OAuth2 credentials stored in n8n. The LLM uses an API key stored in n8n credentials.

9. Additional Deliverables

9.1 API Documentation

The system does not expose its own REST API. Its only public endpoint is the Telegram webhook handled by n8n. It consumes the following external APIs:

API	Operation used	Purpose
Telegram Bot API	Webhook updates (message, callback_query); sendMessage with inline_keyboard or force_reply	Receive patient input, send menus and replies
Google Calendar API	events.list (time range), events.insert (summary, description, start, end), events.delete	Check availability, create and cancel appointments
Google Sheets API	Append row, look up rows, update row	Bookings, ledger, feedback
Gmail API	messages.send	Daily financial report to the doctor
LLM API	Chat completion (system prompt, user message, per-chat memory)	Free-text answers and feedback sentiment

9.2 Testing & Validation Plan

Testing has three levels: **unit** (Code node logic checked with sample inputs), **integration** (n8n executions against a test calendar and test sheets), and **user acceptance** (team members acting as patients on the real bot). The workflow must be active (published) when testing conversation state. Defects are tracked as GitHub Issues.

ID	Test	Level	Expected result
T1	Slot generation on a day with no events	Unit	18 slots from 09:00 to 14:40.
T2	An event 10:00 to 10:20 exists	Unit	The 10:00 slot is removed, others remain.
T3	Today's past slots; Friday and Saturday	Unit	Past slots hidden; Friday and Saturday not offered.
T4	Name and age "[redacted] 25" and Arabic digits	Unit	Valid. Arabic digits are converted.
T5	Name with two words, or age 150	Unit	Invalid; the bot asks again.
T6	Booking with name and age sent as a Reply	UAT	Summary shown; booking created after Confirm.
T7	Booking with name and age sent as a plain message	UAT	Same result as T6 (conversation state).
T8	/start or a menu button in the middle of a booking	UAT	Menu shown; a later text does not create a booking.
T9	Slot taken between display and Confirm	Integration	Bot says the slot is unavailable; no event created.
T10	Free-text question such as the fees	UAT	AI answers from clinic information; no booking created.
T11	Medical question to the AI	UAT	No diagnosis; the patient is redirected to the clinic.
T12	Reminder run twice for the same booking	Integration	Only one reminder sent (reminder_sent).
T13	Cancel from a reminder	Integration	Calendar event deleted; status Cancelled.
T14	Daily report with known ledger rows	Integration	Total and case count equal a manual calculation.
T15	Sample positive and negative feedback texts	Integration	Classified Positive and Negative correctly.
T16	Patient A tries to act on patient B's booking	UAT	Rejected (chat_id mismatch).

9.3 Deployment Strategy

Topic	Plan
Hosting	n8n Cloud (managed, public HTTPS, simplest) or self-hosted n8n (Docker) behind a tunnel or reverse proxy.
Environments and release	Work on a duplicate of the workflow, test, then publish. Workflows are exported as JSON and versioned in GitHub without credentials.
Secrets	Telegram token, Google OAuth, and LLM API key only in n8n credentials. A leaked token is revoked with @BotFather and replaced.
Monitoring	An Error Trigger workflow that sends a Telegram alert to the team when a workflow fails.
Backups	Regular workflow exports and the Google Sheets version history.
Scaling	One doctor means low volume. Limits to watch: Google API quotas and LLM cost. If the clinic grows, move conversation state and bookings to a database such as PostgreSQL.

Appendix — Status and Open Items (6 November 2026)

Item	Status
Booking flow with Calendar availability and the double-check	Working (first version).
Conversation state, so name and age work without replying	Added by the team (reported).
AI Agent for free-text answers	Added by the team (reported). Model and exact prompt to be documented.
Google Sheets logging, reminders, daily report, feedback	In progress (Weeks 5 and 6).
Phone number collection	Not implemented yet (name and age only).
Cancellation policy (3 hours)	Assumption, to be confirmed with the client.

Note: the Project Proposal, README, and Project Info Form still describe the LLM as used only for feedback sentiment. They should be updated to include the AI Agent.